

Enterprise Risk Management — Instructor Guide and Solutions

A mid-sized bank operates a trading desk managing a diversified market-risk portfolio with a total market value of **€100 million**. The bank has implemented an ERM framework centered on quantitative risk limits, with Value-at-Risk (VaR) as the primary risk metric.

How to use this instructor guide

This instructor guide contains full solutions, teaching notes, common student mistakes, and suggested discussion prompts. The structure follows the student workbook but expands each exercise with worked calculations, interpretation, and pedagogical commentary. All data generation and calculations are described using Excel procedures so instructors can reproduce results in class without programming.

Part I — Risk Measurement Foundations (Solutions)

1 Exercise 1.1 — Conceptual VaR (Instructor notes)

Model answer (short):

- $\text{VaR}(1\text{-day}, 99\%) = 10\text{m}$ means: under the model and normal market conditions, there is a 1% probability that the portfolio will lose more than 10m in one day.
- Over 100 trading days, expect about 1 exceedance.
- VaR does not describe the magnitude of losses beyond the threshold; it is a quantile, not an expected tail loss.
- Examples of misleading situations: (1) heavy-tailed returns where extreme losses are much larger than VaR; (2) sudden correlation shifts that increase portfolio risk but are not captured by the VaR model.

Teaching note: Emphasize the difference between frequency (VaR) and severity (ES). Ask students to explain VaR to a non-technical board member in one sentence.

2 Exercise 1.2 — Historical VaR with a realistic sample (Instructor solutions)

Excel dataset generation and sample outputs

Excel steps recap:

1. Column D contains adjusted returns computed as $\text{=NORM.INV(RAND(),0.0004,0.01)} * (1 + 0.5 * \text{ABS(NORM.INV(RAND(),0,0.8)))}$ copied for 250 rows.
2. Inject negative jumps by subtracting 0.05, 0.08 or 0.12 at five chosen rows.
3. Column E contains losses: = -D2 .
4. VaR95: $\text{=PERCENTILE.EXC(E2:E251,0.95)}$.
5. VaR99: $\text{=PERCENTILE.EXC(E2:E251,0.99)}$.
6. ES95: $\text{=AVERAGEIF(E2:E251,">=" cell_with_VaR95)}$.
6. ES99: $\text{=AVERAGEIF(E2:E251,">=" cell_with_VaR99)}$.

Example instructor outputs (representative):

- Historical VaR 95% (loss fraction): 2.4%
- Historical VaR 99% (loss fraction): 4.8%
- Historical ES 95% (loss fraction): 3.5%
- Historical ES 99% (loss fraction): 7.2%
- VaR99 in euros (portfolio 50m): 2.4m
- ES99 in euros (portfolio 50m): 3.6m

Pedagogical commentary:

- Show students how the PERCENTILE and AVERAGEIF functions work and why ES uses an average of tail losses.
- Demonstrate backtesting: count breaches with $\text{=COUNTIF(E2:E251, ">" cell_with_VaR99)}$ and compare to expected 1% = 2.5).
- Discuss sampling variability: repeat the random generation (recalculate) and show how VaR/ES move.

Common student mistakes:

- Using PERCENTILE.INC vs PERCENTILE.EXC without understanding discrete-sample implications.
- Averaging the wrong subset for ES (e.g., averaging returns rather than losses).
- Forgetting to fix random seed equivalents in Excel (use manual values or document the chosen jump rows).

3 Exercise 2.1 — Parametric vs Historical VaR (Instructor solutions)

Solution outline and Excel formulas:

1. Compute sample standard deviation: =STDEV.S(D2:D251).
2. Parametric VaR99: =NORM.S.INV(0.99) * stdev.
3. Compare parametric VaR to historical VaR99 and discuss differences.
4. Stress test: compute recent 20-day volatility =STDEV.S(D232:D251), double it, and recompute parametric VaR.

Teaching note: Use this to show how parametric VaR underestimates tail risk when returns are non-normal.

Part II — Applied ERM Case Studies (Solutions and Teaching Notes)

4 Case 1 solutions: Market risk, VaR limits and stress

Given (restated)

Equity exposure: 60m, equity daily vol 2% $\Rightarrow$ equity daily $\sigma_E = 60,000,000 \times 0.02 = 1.2m$.

Interest-rate exposure: 40m, yield vol 5 bps, sensitivity +1 bp = 0.20m $\Rightarrow$ yield daily $\sigma_{IR} = 0.20m \times 5 = 1.0m$. Correlation = 0.

1. VaR calculation (variance-covariance)

$$\sigma_P = \sqrt{\sigma_E^2 + \sigma_{IR}^2} = \sqrt{1.2^2 + 1.0^2} = 1.561 \text{ m.}$$

$$\text{1-day 99\% VaR (parametric)} = 2.33 \times 1.561 = \mathbf{3.64m.}$$

2. Compliance

Limit = 3.0m. VaR = 3.64m > limit $\Rightarrow$ **Not compliant.**

3. Stress scenario

Equity loss: 60m * 8% = 4.8m.

IR loss: 0.20m per bp * 30 bps = 6.0m.

Total stress loss = 10.8m.

4. Why stress loss can exceed VaR

- VaR is a quantile under normal assumptions; stress scenarios are deliberately outside those assumptions.
- VaR ignores tail severity and assumes stable volatilities and correlations.
- VaR is a one-day metric; stress scenarios can combine multiple drivers and non-linearities.

5. Two complementary ERM controls

1. Liquidity buffer sized by margin-at-risk and a requirement to stress liquidity under extreme scenarios.
2. Scenario limits and concentration limits that trigger mandatory reduction of exposures when breached.

Class activity: Ask students to implement the VaR calculation in Excel and to create a small dashboard showing VaR, limit, and a red/yellow/green indicator.

5 Case 2 solutions: Expected Shortfall, governance and escalation

Data (sorted)

Sorted P&L (worst to best): -6.8, -4.2, -3.5, -2.3, -1.7, -1.1, -0.8, -0.6, +0.4, +0.5, +0.6, +0.9.

1-2. Empirical 1-day 99% ES

With 12 observations, the discrete-sample interpretation places the 1% tail on the single worst observation.
 $ES_{99} = 6.8m$.

3. ES limit breach

Limit = 4.0m. $ES = 6.8m > \text{limit} \Rightarrow$ **Breach**.

4. Possible failures

- Risk reporting buried tail losses or used soft language.
- Governance lacked automatic escalation triggers.
- Incentives rewarded growth and discouraged early escalation.

5. Suggested escalation rule (example)

- Quantitative: $ES_{99} > 75\% \text{ of limit} \Rightarrow$ immediate CRO notification and weekly committee review.
- Qualitative: any single-day loss $> 50\% \text{ of limit} \Rightarrow$ immediate escalation.

6 Case 3 solutions: Signals without escalation

1. ERM framework type

Principles-based, committee-driven, judgment-heavy.

2. ERM architecture

- Risk-takers: business units.
- Risk monitors: central risk function.
- Escalation decision: senior committee (discretionary).

3. Early warning signals

Concentration, optimistic assumptions, frequent exceptions, stress tests near thresholds. They failed to trigger action due to soft language and lack of mandatory triggers.

4. Incentives and culture

Examples: bonuses tied to growth; committees avoiding confrontation.

5. Two governance changes

1. Hard escalation triggers with automatic routing to CRO/CEO.
2. Independent second-line authority with veto power over exceptions.

7 Case 4 solutions: Hedging effectiveness (prospective vs retrospective)

Given

Normal period: $\text{Var}(X)=100$, $\text{Var}(H)=64$, $\text{Cov}=48$. Stress period: $\text{Var}(X)=225$, $\text{Var}(H)=144$, $\text{Cov}=12$.

1. Hedge ratio

$$h^* = \frac{\text{Cov}(X, H)}{\text{Var}(H)} = \frac{48}{64} = 0.75.$$

2. Prospective effectiveness

$$\text{Var}(Y) = 100 + 0.75^2 \cdot 64 - 2 \cdot 0.75 \cdot 48 = 64.$$

$$HE_{pros} = 1 - \frac{64}{100} = 36\%.$$

3. Retrospective effectiveness

$$\text{Var}(Y) = 225 + 0.75^2 \cdot 144 - 2 \cdot 0.75 \cdot 12 = 288.$$

$$HE_{retro} = 1 - \frac{288}{225} = -28\%.$$

4. Why HE differs

Correlation collapse, volatility regime shift, and basis risk.

5. Liquidity and governance

Average margin 18m < buffer 25m (sufficient on average). If spikes to 40m, require contingency lines, margin-at-risk limits, and dynamic hedging rules.

8 Case 5 solutions: When hedging increased risk

1. Main risks

Commodity price, basis, liquidity, funding, operational.

2. Why hedge looked effective ex ante

Historical correlations and variance reduction metrics.

3. Interpretation of h^*

Hedge 90% of exposure assuming stable correlation.

4. Why prospective effectiveness failed

Correlation breakdown, volatility spike, margin calls.

5. Secondary risks ignored

Liquidity, funding, clearing, model risk.

6. ERM governance changes

Liquidity stress tests, limits on margin exposure, contingent credit lines.

9 Case 6 solutions: Shell — Instructor commentary

1. ERM characterization

Principles-based, decision-oriented; scenario-driven.

2. Principal risks philosophy

Focus on material risks; trade-off: blind spots.

3. Scenario analysis

Scenarios test resilience; not forecasts.

4. Architecture and separation

Risk owners in business; central ERM provides challenge.

5. Failure mode

Correlated geopolitical and commodity shock; early warnings: margin calls, basis widening, funding stress.

Part III — Group ERM Design Exercise (Instructor guidance)

Suggested grading rubric (out of 20):

- Coherence of philosophy and scope: 3
- Practicality of governance architecture: 4
- Quality of risk identification and assessment: 4
- Appropriateness of risk responses and instrument governance: 3
- Integration into strategy and capital allocation: 3
- Reporting and escalation design: 2
- Anticipation of failure modes and early warnings: 1

Instructor tips:

- Encourage students to be realistic and implementable.
- Ask groups to present a one-page dashboard and explain each indicator.
- Use the Excel dataset to illustrate how dashboards would have signalled rising risk.

Appendix: Synthetic dataset guidance and sample outputs

Excel guidance: The student workbook instructs how to create a 250-row synthetic return series in Excel and how to compute VaR/ES using PERCENTILE.EXC and AVERAGEIF. For instructor convenience, here are representative sample outputs you can expect when following the steps:

- Historical VaR95%: 2.4% (loss fraction)
- Historical VaR99%: 4.8% (loss fraction)
- Historical ES95%: 3.5% (loss fraction)
- Historical ES99%: 7.2% (loss fraction)
- VaR99 (portfolio 50m): 2.4m
- ES99 (portfolio 50m): 3.6m

Final instructor note: Use the student workbook in class for exercises and this guide to lead discussions. Encourage students to perform the Excel steps themselves and to experiment with alternative stress scenarios (e.g., correlation jumps, liquidity shocks). The goal is to build intuition about how numbers map to governance decisions.

End of instructor guide.